

A Japanese Language Model and Three New Evaluation Benchmarks for Pharmaceutical NLP

JPharmatron & JPharmaBench



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Motivation

The Challenge: LLMs struggle in domain-specific settings requiring deep contextual understanding and precise terminology handling.

- **Japanese pharmaceutical industry** faces significant administrative overhead
- **Gap:** No LLMs or benchmarks for Japanese pharmaceutical operations
- **Medical ≠ Pharmaceutical** - Different terminology, regulations, workflows

Our Contributions

1. **JPharmatron** - First Japanese LLM for pharmaceutical operations via CPT on **2B JP + 8B EN** tokens
2. **JPharmaBench** - First benchmark suite with **3 novel benchmarks** for Japanese pharmaceutical NLP

Building JPharmatron

Base Model: Qwen2.5-7B (Multilingual, commercially permissible)

Training Data: Journal articles, PubMed (~8B tokens), PMDA package inserts, Government docs, FineWeb/Swallow

JPharmaBench: Three Benchmarks

Benchmark	Format	Main Skill	#Examples
YakugakuQA	5-choice QA	Factual recall	3,021
NayoseQA	5-choice QA	Term. normalization	34,769
SogoCheck	Sentence pair	Inconsistency det.	200

YakugakuQA: Japanese National Pharmacist License Exam (2012-2024), 9 categories. Tests factual recall of pharmaceutical knowledge.

NayoseQA: Drug name normalization from KEGG DRUG database. Maps between Japanese↔English and Brand↔Generic names.

SogoCheck: Cross-document consistency detection from PMDA package inserts. Identifies factual discrepancies between related documents.

SogoCheck Example:

Text A: Temp below 25°C, Humidity 60%

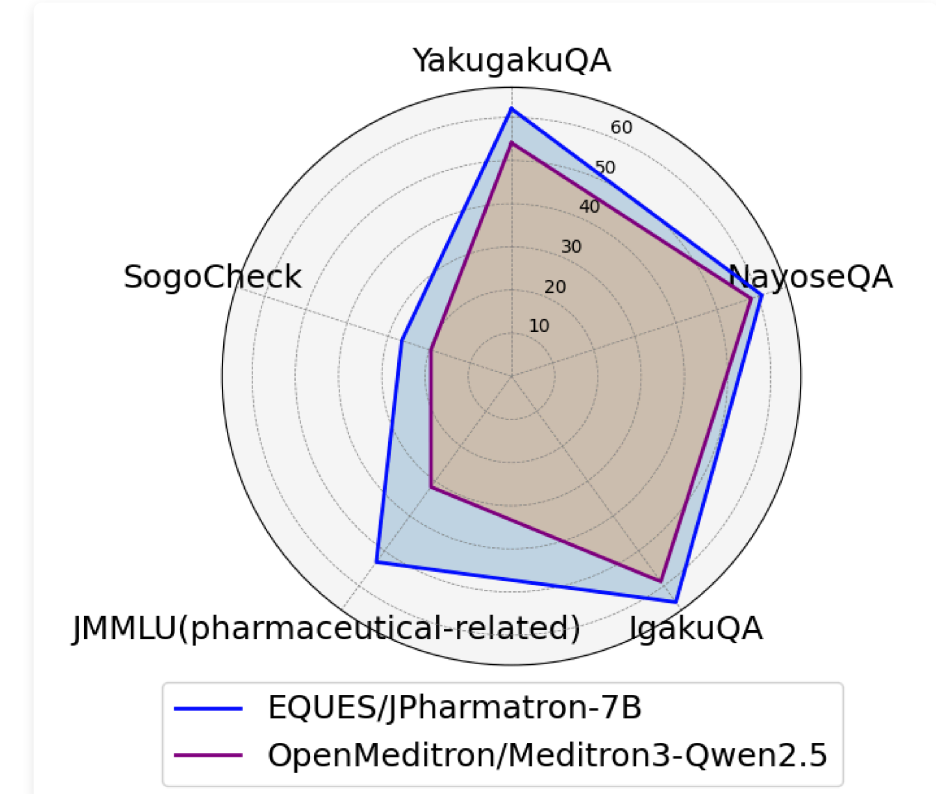
Text B: Temp below 26°C, Humidity 61%

→ **Labels:** Change in temp, Change in humidity

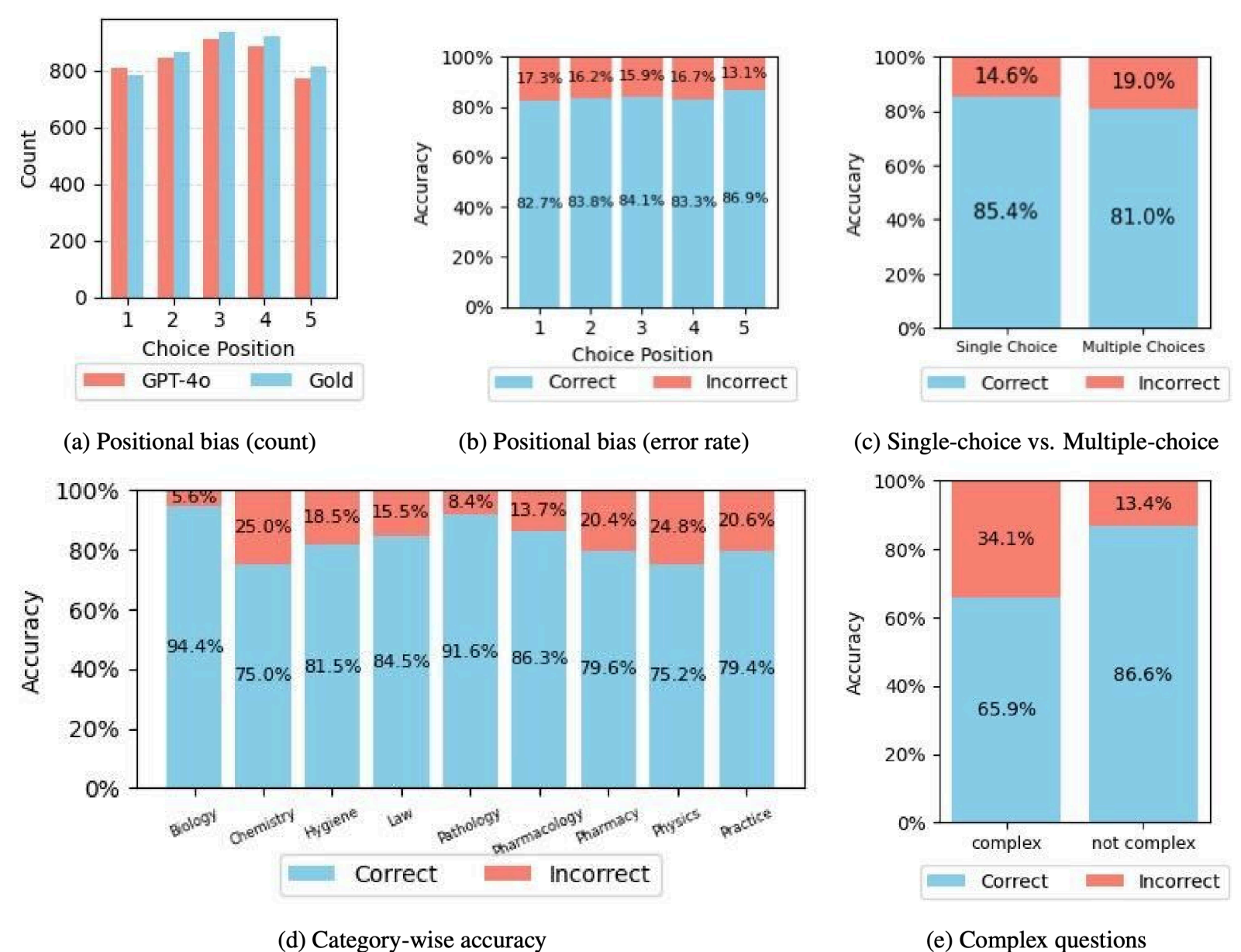
Evaluation Results

Model	YakugakuQA	NayoseQA	SogoCheck	IgakuQA
Meditron3-Qwen2.5-7B	54.1	58.3	19.6	58.8
JPharmatron (best)	62.0	60.9	26.7	64.7
GPT-4o	83.6	86.0	39.1	86.6

Key Insight: Even **GPT-4o** achieves only **39.1%** on SogoCheck! Cross-sentence consistency reasoning remains an **open challenge**.



Error Analysis (GPT-4o on YakugakuQA)



- **Positional bias:** Tendency to favor option "1"
- **Category:** Biology ~94% ✓, Chemistry ~75% ✗
- **Complex questions:** 34.1% error rate

Key Findings

- **+7.9%** improvement over Meditron on YakugakuQA
- Medical specialization alone is **insufficient** for pharmaceutical tasks
- SogoCheck is **challenging for ALL models**
- Top-tier LLMs still struggle with **calculation-intensive** tasks

Conclusion

- **First** Japanese pharmaceutical LLM and benchmark suite
- Demonstrates **cost-effective, privacy-conscious** domain adaptation
- **SogoCheck** reveals open challenges in cross-document consistency